

John Leonardo

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PROFESSIONAL SUMMARY Senior Software Engineer, 6+ years in production ML/AI and distributed infrastructure, 80k TPS search to agent platforms. Architect of a production LLM agent and its evaluation system; owner of the kernel-enforced security boundary for Roblox's company-wide AI coding agents.

EMPLOYMENT

Senior Software Engineer Roblox June 2025 - Present Architected the production AI assistant backend: supervisor routing three LangGraph subgraphs, ReAct tool-calling bounded at 10 iterations and 40 node executions with rolling summarization, an enumerated 19-operation tool surface instead of a full API, and per-user delegated auth via short-lived transaction tokens with identity injected server-side into agent runtime config, never prompt content. Designed the retrieval eval system from first principles after inheriting it off-track and pulled from CI for a security finding: replaced synthetic personas scored by an LLM judge, uncorrelated with human raters, with deterministic document-retrieval scoring against golden ground truth. Tool-call recall 32% to 62% on a 52-builder gold set, no regression across IC/manager, job family, or comp-level subgroups. Own agent sandbox and containment for Roblox's company-wide AI coding agent rollout: the kernel-enforced network and filesystem boundary, default-installed on every Roblox Mac, that all engineer-run AI agents execute inside. Hardened kernel network enforcement in a macOS System Extension (transparent-proxy interception, TLS inspection, selective MITM) traced a four-stage path silently blocking internal endpoints, shipped a port-enforced private-CIDR exemption rejecting prefixes straddling public space, mutation-tested the guard to prove exactly two of 1,179 tests gate it. Designed multi-app generalization of the enforcement daemon: EndpointSecurity AUTH_EXTENSION classification by code-signing identity for race-free registration, replacing per-app wrapper coupling with a two-tier trust model and per-TLS-stack delivery via MDM. Introduced Temporal; built the async platform every background AI feature runs on. Self-reconciling schedules, ~10 lines per new job type, 3,000+ users per cycle inside an 8-hour overnight window. Killed pool exhaustion and silent model fallbacks under unbounded agent fan-out: capped concurrency at 30/pod (150 cluster-wide, matched to the DB pool) and 20 TPS against a 30 TPS gateway budget. Sustains ~1,520 jobs and 4,091 tool-calling LLM calls per run.

Software Engineer II Amazon September 2021 - June 2025 Delivered and deployed a RoBERTa-based NER Query Understanding solution using an ensemble model architecture: split ONNX for encoding and NER predictions. Made use of Triton+ ECS and launched A/B experiment which confirmed \$290MM additional revenue for Amazon. Key contributor to the launch of Rufus, Amazon's LLM-based shopping assistant, by developing a Query Planner service. Productionized quantized RoBERTa models (Triton ECS) enabling 20k TPS processing while maintaining detent height error rate below 0.03%. Upgraded Amazon Search service infrastructure (80k TPS) implementing AppConfig-based throttling, migrating logs to AWS CloudWatch, modernized to JDK 17, and transitioned systems from a legacy system (Apollo) to AWS within six months. Directly managed 3 interns and mentored 4 engineers, with 3 successful SDE1->SDE2 promotions (including promo document authorship).

Software Engineer IBM August 2020 - August 2021 Built a Salesforce-to-OpenShift integration for batch processing, increasing throughput by 300% for 10GB+ daily data volume. Built a RESTful abstraction layer over legacy print infrastructure, cutting processing time from 4 minutes to 20 seconds.

EDUCATION Sacramento, CA California State University, Sacramento August 2017 - May 2020 B.S. in Computer Science with Minor in Mathematics

PROJECTS
resumerodeo - AI resume coaching (Gemini+ Next.js); 1000+ users. **adaptivewealth.ai** - AI financial advisory. Co-founder(departed); raised \$500K seed at \$7.1M valuation, AI2 Incubator.
stockquantai.com - AI agent for quantitative research in US equities. **link.fyi** - Linkshortener that just works.
jdleo.me/blog & jdleo.me/apps - 12+ technical articles and apps on LLMs and AI architectures.

SKILLS
Languages: Python, TypeScript/JavaScript, Java, Go, Swift, Kotlin, C++, Rust, C#, Elixir
Frameworks & Libraries: React, Next.js, Spring Boot, FastAPI, Flask, Node.js, Express, GraphQL, Tailwind CSS, SwiftUI, Temporal, Playwright, Selenium, Ktor, D3.js, Recharts, Redux, Zod, Bun, Gradle, CMake, Bazel, HealthKit, StoreKit
Cloud, Infra, Databases: AWS (S3, Lambda, ECS, CloudWatch, DynamoDB), GCP, Firebase, Vercel, Kubernetes, Docker, OpenShift, Terraform, GitLab CI/CD, PostgreSQL, MongoDB, Redis, SQLite, Firestore, Neon; Datadog, CloudFormation
ML/AI: PyTorch, scikit-learn, XGBoost, LightGBM, TensorFlow, ONNX, Triton; fine-tuning transformers (RoBERTa, BERT); RAG, knowledge graphs, embeddings, semantic search; LLM integrations (Claude, GPT, Gemini); A/B testing, NumPy, Pandas